

ComicCraft AI

Problem-Solution Fit, Proposed Solution & Solution Architecture

1. Problem-Solution Fit Canvas

Category	Details
1. Customer Segment (CS)	Creative writers, students, and independent comic enthusiasts.
2. Problems / Pains (PR)	Inability to draw comic panels; professional software is too hard to learn.
3. Triggers to Act (TR)	Having a great story idea and wanting to publish it visually.
4. Emotions Before/After (EM)	Before: Frustrated and restricted. After: Empowered, satisfied, and happy.
5. Available Solutions (AS)	Hiring professional artists (Expensive) or using complex drawing software (Steep learning curve).
6. Customer Limitations (CL)	Zero or low budget for illustration, lack of digital art skills.
7. Behavior (BE)	Writing stories in notebooks/Word docs and searching for free AI image generators online.
8. Channels of Behavior (CH)	Online (Web browsers, AI tool websites).
9. Problem Root Cause (RC)	Lack of artistic abilities and the high cost/time required for comic production.
10. Your Solution (SL)	ComicCraft AI - A web app that automatically generates comic story dialogues and images from simple text prompts, exporting them as PDFs.

2. Proposed Solution Report

Project Overview

Objective: The primary objective is to automate and simplify the comic creation process by implementing generative AI techniques, ensuring fast and visually appealing story panels.

Scope: The project provides a web-based platform that processes user text prompts, generates stories and matching images, and compiles them into a downloadable PDF format.

Problem Statement

Description: Addressing the inability of non-artistic writers to visualize their stories, which limits their creative expression and content sharing.

Impact: Solving these issues will result in an easy, zero-cost way for anyone to create comics, contributing to enhanced digital storytelling and customer satisfaction.

Proposed Solution

Approach: Employing Google Gemini and Stable Diffusion APIs to dynamically generate text and images based on user inputs.

Key Features:

- Implementation of an AI-based text and image generation model.
- Real-time preview of comic panels.
- Automated PDF compilation and export feature.

Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	Standard CPU for web app execution	Intel i5 / Ryzen 5 or equivalent
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data and local image files	20 GB Free Space
Frameworks	Python web frameworks	FastAPI
Libraries	Additional libraries	FPDF, Jinja2, Requests
Development Environment	IDE	VS Code or PyCharm

3. Solution Architecture

Architecture Diagram Flow (Text Representation)

Presentation / Client Layer	API Gateway / Server	Core Logic Service	External API Integrations	Data / Storage Layer
Web App HTML5, CSS3, Jinja2 Templates	Uvicorn / FastAPI Server	Prompt Processing PDF Compilation (FPDF)	Google Gemini API (Text) Stable Diffusion API (Images)	Local File System Static folder for panels and PDFs

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the web interface for users to enter prompts and view the generated comic.	HTML, CSS, Jinja2
Application Server	Routes user requests, handles asynchronous API calls, and processes data.	Python, FastAPI
External APIs	Acts as the brain of the app, generating the storyline dialogues and the visual panel artworks.	Gemini API, Stable Diffusion API
PDF Exporter	Takes the generated text and images and formats them into a final downloadable document.	Python FPDF Library
Data Storage	Temporarily holds the downloaded API images and final PDFs before serving them to the user.	Local File System